PROJECT-BASED LEARNING

PHASE-I EVALUATION

Emergency Call Detection & Priority Routing

Team ID: T010

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team ID: **T010**
Semester: ***3rd SEM***
Domain: ***DS/CPP***

Team Members

1. *Avika Dixit – 2028404*
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Mentor

Ms. Priyanka Rastogi, Assistant Professor, CSE Dept **Interactions completed: 3**

Problem Statement & Motivation

Problem Statement

Emergency Services may receive multiple duplicate, incomplete or conflicting reports about the same incidents, causing confusion and unnecessary delay in handling crisis emergencies.

Motivation

- Duplicate Reports can overload emergency operators
- Critical Incidents needs to identified quickly
- Pranks call should be identified.
- Existing focus on receving and dispatching the call not prioritize it.

Target Users / Stakeholders

- **Primary:** Emergency operators and Dispatchers.
- **Secondary:** Police, Fire, Medical Rescue Team.
- **Benefit:** Faster Incident Identification, Better Prioritization, Better response team dispatch

Objectives

Project Objectives

1. Implement an incoming report ingestion system using FIFO Queues.
2. Build a Call Analyser to detect low-information or fake calls.
3. Construct report graphs and group related reports via Disjoint Set Union (DSU).
4. Prioritise incidents dynamically using Priority Queues for fast dispatch
5. Integrate advanced Data Structures with Object-Oriented Design.

Key Objective: **To group related emergency reports iinto one incident and give priority to important incidents for faster response**

Proposed Solution

Proposed Approach

1. **Queue Manager:** FIFO order storage.
2. **Call Analyzer:** Computes similarity /quality.
3. **Relationship Builder:** Hash Maps and Graphs.
4. **Incident Manager:** DSU clustering.
5. **Dispatcher:** High-priority routing

Key Features

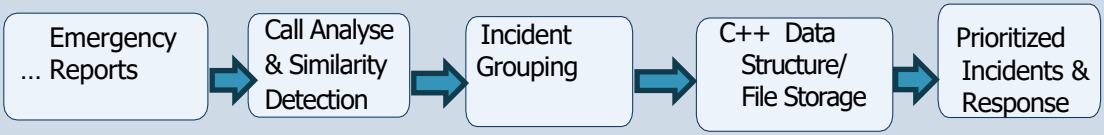
- ▶ Automated Fake Call Detection.
- ▶ Fast O(1) Hash Map Lookups.
- ▶ Dynamic DSU Incident Grouping.
- ▶ Priority-based Dispatching.
- ▶ Feedback Learning Loop.

The System will group related emergency reports, identify important incidents, prank call and prioritise then for faster team allocation at right place

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	**Queue,Hashing,Graph,DSU,Priority Queue**	Report processing ,grouping , and Prioritisation
OOPs with C++	**Classes, Inheritance, Polymorphism, STL.**	Call, Incident, Disatcher and Response Team Module

System Architecture / Workflow



Major Components

- Emergency Call Management
- Call Analysis & Incident Grouping
- Priority & Response Managing

Data / Control Flow

- Reports are analysis , grouped, prioritised and then responded.
- Each module process data and passes it to next.
- **Output**: Grouped incidents,priority, and response team

Technology Stack & Methodology

Technology Stack

- Language: C++
- Database: File-based storage
- IDE: VS Code
- Version Control: Git & GitHub

Development Methodology

1. Identify the problem and define system requirement.
2. System Architecture and workflow
3. Develop core module/classes using C++
4. Test the system with different situations
5. Analyze the output and overall performance

Justification:

- C++ provide direct implementation of DSA,STL and OOPS concepts.
- Github provide a setup for development and collaboration.
- The Methodology ensue development and testing evaluations

Initial Progress & Team Contribution

Progress Achieved

- To identify duplicate and related emergency reports.
- Studied existing systems and their limitation.
- Identified how required system will function.
- Prepared system Architecture.
- Designed the workflow of the system

Individual Contribution

Member	Role	Contribution
Avika Dixit Queue	Lead	Architecture
Saksham DSU	Developer	Analyzer
Nitin Bhandari Graph	Backend	Hash Map
Harshita Docs	Testing	Priority Queue

Roadmap, Expected Outcomes & References

Roadmap

1. **Phase-I:** Problem Analysis , Project Proposal ,Design, Workflow
2. **Phase-II:** Develop and implement of main features.
3. **Phase-III:** Testing of final system.

Expected Outcomes

- System to identify and group same emergency reports
- To improve incident Prioritisation System.
- Can be expand in futurewith advance features

References

1. Bjarne Stroustrup, The C++ Programming Language , Addison Wesley.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms , MIT Press.
3. GeeksforGeeks – Data Structures and Algorithms , C++ and Graph/DSU concepts.a
4. cppreference — C++ language and Standard Library reference.
5. Materials related to the course lectures and laboratory of the Data Structures and Object-Oriented Programming .

Thank You
Questions & Suggestions